

0.1 Reduced densities and masses of repeated pure states

Let $\mathcal{H}_S = \mathbb{C}^S$ and $\mathcal{H}_R = \mathbb{C}^\Omega$ be finite-dimensional, and let $\psi \in \mathcal{H}_S \otimes \mathcal{H}_R$. Write $\rho = \text{tr}_S |\psi\rangle\langle\psi|$ for its actual reduced density, without imposing normalization. For $k \geq 0$, regroup the repeated vector into the two systems. Its coordinates, indexed by $x \in S^k$ and $y \in \Omega^k$, are

$$v_k(x, y) = \prod_{i=0}^{k-1} \psi(x_i, y_i). \quad (1)$$

These are the pure-state identities used in the Schur-sector argument of [OAI26, Section 7].

Theorem 0.1.1 (Reduced density of a repeated bipartite vector). Lean ✓ Stmt For every ψ and every $k \geq 0$,

$$\text{tr}_{S^k} |v_k\rangle\langle v_k| = \rho^{\otimes k}. \quad (2)$$

Proof. ✓ Proof For $y, z \in \Omega^k$, the matrix entry on the left is

$$\sum_{x \in S^k} \prod_i \psi(x_i, y_i) \overline{\psi(x_i, z_i)} = \prod_i \sum_{s \in S} \psi(s, y_i) \overline{\psi(s, z_i)} = \prod_i \rho(y_i, z_i). \quad (3)$$

The middle equality is the distributive law for a finite product of sums. □

Theorem 0.1.2 (Projection mass of a repeated bipartite vector). Lean ✓ Stmt For every orthogonal projector P on $\mathcal{H}_R^{\otimes k}$,

$$\|(I_{S^k} \otimes P)v_k\|^2 = \text{Re tr}(\rho^{\otimes k} P). \quad (4)$$

In particular, taking P to be an actual central Schur projector gives the squared norm in that sector.

Proof. ✓ Proof The trace-pairing property of the partial trace and the preceding identity give

$$\text{tr}(\rho^{\otimes k} P) = \langle v_k, (I_{S^k} \otimes P)v_k \rangle = \langle (I_{S^k} \otimes P)v_k, (I_{S^k} \otimes P)v_k \rangle. \quad (5)$$

The second equality uses $P = P^\dagger = P^2$. Taking real parts proves the formula. □

Both identities include zero copies and empty one-copy factors; no invertibility of ρ is required. Concentration, the entropy window and selection of a common sequence of labels are separate assertions.

0.2 A common sequence of high-mass entropy labels

Let $q = \dim \mathcal{H}_R$ and $N_k = (k+1)^{q^2}$. The permutation action on $\mathcal{H}_R^{\otimes k}$ has central projectors Π_λ , indexed by irreducible labels λ of the symmetric group. Write d_λ for the irreducible dimension. The following statements give the simultaneous selection in [OAI26, Section 7], equation (comparator:high-label).

Theorem 0.2.1 (Finite entropy-window criterion). Lean ✓ Stmt Let σ be a density matrix on $\mathcal{H}_R^{\otimes k}$ which commutes with all copy permutations. Let $a \in \mathbb{R}$ and $w \geq 0$. Set $L = -\log \sigma$, with the logarithm assigned value zero on the kernel. If

$$\text{tr}(\sigma 1_{\{|L-a|>w\}}) + N_k e^{-w/2} \leq \frac{1}{2}, \quad (6)$$

then there is a label λ such that

$$\text{tr}(\sigma \Pi_\lambda) \geq \frac{1}{2N_k}, \quad |\log d_\lambda - a| \leq 2w. \quad (7)$$

Proof. ✓ Proof Resolve σ and the central label projectors simultaneously. On a joint eigenspace of positive density eigenvalue u and label λ , put $\ell = -\log u$ and $f = \log d_\lambda$. The eigenvalue multiplicity bound $ud_\lambda \leq 1$ implies $f \leq \ell$. If $|f - a| > 2w$, then either $|\ell - a| > w$ or $\ell - f > w$. The mass of the second event is bounded by $e^{-w/2}$ times the exponential moment of $(L - F)/2$, which is at most N_k . The hypothesis therefore leaves at least half of the total mass in the good labels. At most N_k labels occur, so one good label has mass at least $(2N_k)^{-1}$. Zero eigenvalues contribute no mass. $\square$

Theorem 0.2.2 (Independent-copy entropy window). Lean ✓ Stmt For every density matrix ρ on $\mathcal{H}_R$, for all sufficiently large k there is a label λ satisfying

$$\mathrm{tr}(\rho^{\otimes k} \Pi_\lambda) \geq \frac{1}{2N_k}, \quad |\log d_\lambda - kS(\rho)| \leq 2k^{3/4}. \quad (8)$$

No invertibility of ρ is assumed.

Proof. ✓ Proof The independent-copy surprisal has mean $kS(\rho)$ and variance kV , where V is its one-copy variance. Chebyshev's inequality at width $k^{3/4}$ bounds the tail by $V/\sqrt{k}$. The sum $V/\sqrt{k} + N_k e^{-k^{3/4}/2}$ tends to zero. The finite criterion applies for all sufficiently large k . $\square$

Theorem 0.2.3 (One sequence for mass and entropy). Lean ✓ Stmt For every density matrix ρ , there is one sequence $(\lambda_k)_{k \geq 0}$ such that, for all sufficiently large k ,

$$\mathrm{tr}(\rho^{\otimes k} \Pi_{\lambda_k}) \geq \frac{1}{2N_k}, \quad (9)$$

and

$$\log d_{\lambda_k} = kS(\rho) + o(k). \quad (10)$$

Proof. ✓ Proof Choose a label satisfying both conclusions of the preceding theorem for each sufficiently large k ; at the remaining finitely many values choose any label. The normalized dimension error is at most $2k^{-1/4}$ and therefore tends to zero. $\square$

Theorem 0.2.4 (High-mass labels of actual pure tensor powers). Lean ✓ Stmt Let $\psi \in \mathcal{H}_S \otimes \mathcal{H}_R$ be a unit vector, let $\rho = \mathrm{tr}_S |\psi\rangle\langle\psi|$, and let v_k be its literal tensor power regrouped into $\mathcal{H}_S^{\otimes k} \otimes \mathcal{H}_R^{\otimes k}$. There is one sequence (λ_k) for which, for all sufficiently large k ,

$$\|(I \otimes \Pi_{\lambda_k})v_k\|^2 \geq \frac{1}{2N_k}, \quad (11)$$

and $\log d_{\lambda_k} = kS(\rho) + o(k)$.

Proof. ✓ Proof The actual reduced density is positive semidefinite and has trace one. Apply the common-sequence theorem to this density. The projection-mass identity for actual pure tensor powers converts its trace mass into the stated squared norm. $\square$

0.3 The initial vector with common auxiliary labels

Let $\Omega \in \mathbb{C}^A$ and $\psi \in \mathbb{C}^S \otimes \mathbb{C}^d$ be unit vectors. Suppose that $H\Omega = E_0\Omega$, with $E_0 \in \mathbb{R}$. Write $\rho = \mathrm{tr}_S |\psi\rangle\langle\psi|$. The initial vector below is the construction in [OAI26, Section 7], equations (comparator:prevector) and (comparator:high-label).

Definition 0.3.1 (Initial vector in product coordinates). Lean ✓ Stmt Let $u_d = d^{-1/2} \sum_{i=0}^{d-1} |i, i\rangle$ be the uniform auxiliary pair. Regroup its literal tensor power into the two auxiliary systems and write it as U_k . For a central label projector Π_λ on $(\mathbb{C}^d)^{\otimes k}$, define

$$v_{k,\lambda} = \Omega^{\otimes k} \otimes (I_C \otimes \Pi_\lambda) U_k. \quad (12)$$

All products and projections here act on the actual finite copy spaces.

Theorem 0.3.2 (One sequence for the initial vector and selected mass). Lean ✓ Stmt *There is one sequence (λ_k) for which every $k \geq 1$ satisfies*

$$0 < \|v_{k,\lambda_k}\| \leq 1, \quad (\bar{H}_k \otimes I_{CR}) v_{k,\lambda_k} = E_0 v_{k,\lambda_k}, \quad \bar{H}_k = \frac{1}{k} \sum_{i=0}^{k-1} H^{(i)}. \quad (13)$$

The vector has label λ_k on both auxiliary systems and is invariant under every simultaneous copy permutation on the physical, left auxiliary and right auxiliary systems. For all sufficiently large k ,

$$\|(I_{S^k} \otimes \Pi_{\lambda_k}) \psi^{\otimes k}\|^2 \geq \frac{1}{2(k+1)^{d^2}}, \quad (14)$$

and $\log d_{\lambda_k} = kS(\rho) + o(k)$. The initial projection of U_k is required only to be nonzero; no inverse polynomial lower bound on its probability is asserted.

Proof. ✓ Proof Start with the common sequence selected from the actual density ρ . For any remaining copy number at which the selected sector has zero mass, choose a sector of positive mass using the complete central resolution of $\rho^{\otimes k}$. The original sequence has positive mass eventually, so these changes affect only finitely many terms and preserve the mass bound and entropy asymptotic.

Unit normalization of ψ implies $d > 0$. Every selected projector is nonzero, and hence acts nontrivially on the uniform pair. Tensoring with $\Omega^{\otimes k}$ preserves its norm. Orthogonal projection is contractive, and both unprojected tensor powers have norm one, giving the asserted norm bounds. The one-copy eigenvector equation gives the mean-energy equation on the physical product, irrespective of the auxiliary vector.

The repeated uniform pair has matching left and right central labels. Idempotence therefore gives both fixed-label equations after projection. The uniform pair and the physical product are invariant under simultaneous copy permutations; centrality of the right label projector preserves this invariance after projection. $\square$

Bibliography

- [OAI26] OpenAI, *A two-dimensional area law from a global spectral gap*, September 24, 2026, Section 7.